Edge AI System for Waste Object Analysis Using Fixed-Camera Image Processing

1. Background and Motivation

Waste classification and measurement are key challenges in smart waste management, recycling, and environmental monitoring.
Traditional solutions rely on manual sorting or cloud-based image analysis, which are costly, slow, and require network connectivity.
This project aims to develop an offline, low-cost, AI-at-the-edge system capable of estimating the material, weight, surface area, and quantity of waste or grouped objects using a fixed camera. The system will be optimized for the NVIDIA Jetson Nano and will leverage modern depth estimation models like Depth-Anything for real-time performance.

2. Problem Definition and Objectives

The project seeks to design and implement a system that processes images captured by a fixed camera to infer:
- The material (plastic, glass, metal, paper, etc.)
- The weight estimation (based on dimensions and visual density)
- The quantity of distinct objects
- The surface area and approximate dimensions of each object

Key objectives:
1. Achieve real-time inference on the Jetson Nano.
2. Operate completely offline (no cloud dependencies).
3. Provide scalable and low-cost deployment for IoT or industrial environments.

3. Literature Review

Depth estimation and material recognition from RGB images are active research areas. Recent advancements include monocular depth estimation models such as Depth-Anything (Lihe Young et al., 2024), which generate dense, accurate depth maps from single RGB images. These methods remove the need for depth sensors, reducing costs and complexity.

Other relevant works include:
- MiDaS: multi-scale depth estimation for real-world scenarios.
- Segment Anything (SAM): general-purpose segmentation model.
- YOLOv8/YOLOv9: real-time object detection models suitable for embedded devices.

For material classification, CNN and Vision Transformer models pre-trained on datasets like MINC-2500 can be fine-tuned for material recognition.

4. Proposed System Architecture

The proposed system consists of four main components:
1. Image acquisition module (fixed camera setup).
2. Object detection and segmentation module (YOLOv8-seg).
3. Depth estimation module (Depth-Anything).
4. Feature extraction and weight estimation module.

Hardware:
- NVIDIA Jetson Nano (4GB)
- USB RGB Webcam or Stereo Camera (OAK-D Lite optional)
- Power supply, SD card, and stable mount

Software stack:
- Python, OpenCV, PyTorch
- Depth-Anything / MiDaS for depth inference
- Ultralytics YOLOv8 for detection and segmentation
- Custom post-processing for surface, volume, and weight estimation

5. Methodology

1. Calibrate the camera using an ArUco marker for real-world scaling.
2. Detect objects and extract masks using YOLOv8-seg.
3. Estimate the depth map using Depth-Anything.
4. Compute per-object volume via 3D reconstruction or depth integration.
5. Estimate material type using a lightweight CNN.
6. Derive approximate weight = Volume × Density (ρ).

6. Experimental Design

- Dataset: custom dataset of waste items (bottles, cans, papers, plastics).
- Evaluation metrics: mAP for detection, RMSE for depth accuracy, percentage error for weight estimation.

- Jetson Nano optimizations: TensorRT quantization, model pruning, and mixed-precision inference.
- Validation: compare estimated weights and dimensions to physical measurements.

7. Expected Results

The system is expected to achieve:
- Real-time inference (~10–15 FPS) on Jetson Nano.
- <15% error in weight estimation for most common waste items.
- Robust offline operation under stable lighting conditions.

8. Cost and Alternatives

The proposed setup costs under €200:
- Jetson Nano (~€100)
- RGB camera (~€40)
- Power and peripherals (~€30)

Optional upgrades:
- Luxonis OAK-D Lite (~€200): native depth sensing.
- Intel RealSense D455 (~€250): higher accuracy for research purposes.

9. Repository Structure

Suggested GitHub layout:

```
.
├── data/              # Datasets and calibration images
├── src/
│   ├── detection.py       # YOLOv8 inference
│   ├── depth_estimation.py# Depth-Anything integration
│   ├── material_class.py  # Material classifier
│   ├── weight_estimator.py# Volume & density-based weight
│   └── main.py          # Main pipeline
├── models/            # Pretrained and fine-tuned models
├── notebooks/          # Experiments and visualization
└── docs/              # Project report and diagrams
```

10. Conclusion

This project demonstrates a practical, real-time, and offline AI-at-the-edge solution for smart waste analysis.

It leverages modern monocular depth estimation and segmentation models to estimate physical properties like
dimensions and weight without expensive sensors. The outcome could contribute to scalable waste monitoring,
recycling automation, and industrial IoT applications.